Generalised Langevin Equation

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I

For a harmonic oscillator,

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2 \tag{1.1}$$

the time-evolution of position is

$$\ddot{q} = -\omega^2 q \tag{1.2}$$

The Kubo-transformed real time correlation function between two observables is defined as

$$C_{AB}(t) \equiv \frac{1}{\beta}\int_0^\beta d\lambda \langle A(-i\hbar\lambda)B(t)\rangle \tag{1.3}$$

where

$$\langle \cdots \rangle \equiv \frac{\mathrm{tr}\left[e^{-\beta H}\cdots\right]}{\mathrm{tr}\left[e^{-\beta H}\right]} \tag{1.4}$$

and

$$B(\tau) \equiv e^{+iH\tau/\hbar}Be^{-iH\tau/\hbar} \tag{1.5}$$

Since both differentiation and Kubo-transform are linear, it follows that

$$\ddot{C}_{qq}(t) + \omega^2 C_{qq}(t) = 0 \tag{1.6}$$

Let us now define an inner product using the Kubo-transformed real time correlation function at time zero

$$(A|B) \equiv \frac{1}{\beta}\int_0^\beta d\lambda \, \langle A^\dagger(-i\hbar\lambda)B\rangle \tag{1.7}$$

$$(A|B(t)) = C_{AB}(t) \tag{1.8}$$

For two general non-Hermitian operator

$$(A|B) = (B|A)^* \tag{1.9}$$

Now consider a case of a potential with anharmonicity

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2 + U(q) \tag{1.10}$$

then we have

$$\ddot{q} = -\omega^2 q - \frac{1}{m}U'(q) \tag{1.11}$$

where $U' = \partial U/\partial q$, projecting into position operator at time zero yields

$$\ddot{C}_{qq}(t) + \frac{1}{m}C_{qU'}(t) + \omega^2 C_{qq}(t) = 0 \tag{1.12}$$

The time evolution of an observable in the Heisenberg picture is

$$i\hbar\dot{A} = [A, H] \tag{1.13}$$

we now define the Liouvillian super-operator as

$$\mathcal{L}(\cdot) \equiv \frac{1}{\hbar}[H, \cdot] \tag{1.14}$$

note that all super-operators are denoted by cursive font throughout.
thus we have

$$|\dot{A}(t)) = i\mathcal{L}|A(t)) \tag{1.15}$$

Which is analogous to Schrödinger equation for quantum state.
The formal solution is given by

$$|A(t)) = e^{i\mathcal{L}t}|A(0)) = e^{i\mathcal{L}t}|A) \tag{1.16}$$

We define a projection $\mathcal{P}$ operator as

$$\mathcal{P} \equiv \frac{|A)(A|}{(A|A)} \tag{1.17}$$

it can be shown that the projection operator is idempotent

$$\mathcal{P}^2 = \mathcal{P} \tag{1.18}$$

We define the projection into the orthogonal subspace as

$$\mathcal{Q} \equiv 1 - \mathcal{P} \tag{1.19}$$

It also can be shown that it is idempotent

$$\mathcal{Q}^2 = \mathcal{Q} \tag{1.20}$$

and satisfy the following properties

$$\mathcal{P}\mathcal{Q} = \mathcal{Q}\mathcal{P} = 0 \tag{1.21}$$

$$(\mathcal{P}A|B) = (A|\mathcal{P}B) \tag{1.22}$$

$$(\mathcal{Q}A|B) = (A|\mathcal{Q}B) \tag{1.23}$$

thus these Hermian

The Liouvillian is also Hermitian (Proof: Appendix A)

$$(\mathcal{L}A|B) = (A|\mathcal{L}B) \tag{1.24}$$

which can be shown by using the definition of the Liouvillian, inner product and the cyclic permutation invariance property of trace.

To proceed, consider the Laplace transform

$$\int_0^\infty dt\, e^{-st} e^{-Rt} = (s+R)^{-1} \tag{1.25}$$

and also note that

$$(U+V)^{-1} = U^{-1} - U^{-1}V(U+V)^{-1} \tag{1.26}$$

$$(s+R+S)^{-1} = (s+R)^{-1} - (s+R)^{-1}S(s+R+S)^{-1} \tag{1.27}$$

Using the inverse Laplace transform and the Laplace convolution theorem yields our desired operator identity

$$e^{-(R+S)t} = e^{-Rt} - \int_0^t d\tau\, e^{-R\tau} S e^{-(R+S)(t-\tau)} \tag{1.28}$$

we now use the substitutions

$$R = i\mathcal{L}\mathcal{Q} \tag{1.29}$$

$$S = i\mathcal{L}\mathcal{P} \tag{1.30}$$

to obtain

$$e^{-i\mathcal{L}t} = e^{-i\mathcal{L}\mathcal{Q}t} - i\int_0^t d\tau\, e^{-i\mathcal{L}\mathcal{Q}\tau}\mathcal{L}\mathcal{P}e^{-i\mathcal{L}(t-\tau)} \tag{1.31}$$

where we have used $\mathcal{P} + \mathcal{Q} = 1$, and finally taking the adjoint of both sides yields the desired form of the propagator

$$e^{i\mathcal{L}t} = e^{i\mathcal{Q}\mathcal{L}t} + i\int_0^t d\tau\, e^{i\mathcal{L}(t-\tau)}\mathcal{P}\mathcal{L}e^{i\mathcal{Q}\mathcal{L}\tau} \tag{1.32}$$

II

It follows that

$$|\dot{A}(t)) = ie^{i\mathcal{L}t}\mathcal{L}|A) = ie^{i\mathcal{L}t}(\mathcal{P}+\mathcal{Q})\mathcal{L}|A) \tag{2.1}$$

Now we consider the two projections seperately,

$$ie^{i\mathcal{L}t}\mathcal{P}\mathcal{L}|A) = i\frac{(A|\mathcal{L}A)}{(A|A)}e^{i\mathcal{L}t}|A) = i\Omega|A(t)) \tag{2.2}$$

where we have defined

$$\Omega \equiv \frac{(A|\mathcal{L}A)}{(A|A)} \tag{2.3}$$

which is real, due to $\mathcal{L} = \mathcal{L}^\dagger$.
We now define

$$|\delta F(t)) \equiv ie^{i\mathcal{Q}\mathcal{L}t}\mathcal{Q}\mathcal{L}|A) \tag{2.4}$$

at $t = 0$ we have

$$|\delta F) = i\mathcal{Q}\mathcal{L}|A) \tag{2.5}$$

Applying the propagator and using Eq. (1.31) yields

$$e^{i\mathcal{L}t}|\delta F) = e^{i\mathcal{L}\mathcal{Q}t}|\delta F) + i\int_0^t d\tau\, e^{i\mathcal{L}(t-\tau)}\mathcal{P}\mathcal{L}e^{i\mathcal{L}\mathcal{Q}\tau}|\delta F) \tag{2.6}$$

$$e^{i\mathcal{L}t}|\delta F) = |\delta F(t)) + i\int_0^t d\tau\, e^{i\mathcal{L}(t-\tau)}\mathcal{P}\mathcal{L}|\delta F(\tau)) \tag{2.7}$$

Using the projection operator

$$\mathcal{P}\mathcal{L}|\delta F(\tau)) = \frac{(A|\mathcal{L}\delta F(\tau))}{(A|A)}|A) = \frac{(\mathcal{L}A|\delta F(\tau))}{(A|A)}|A) \tag{2.8}$$

Then notice that

$$\mathcal{P}|\delta F) = 0 \tag{2.9}$$

furthermore, we have

$$\mathcal{P}|\delta F(\tau) = 0 \tag{2.10}$$

as all the terms in the Taylor series expansion of $e^{i\mathcal{Q}\mathcal{L}\tau}$ starts off with a $\mathcal{Q}$ and $\mathcal{P}\mathcal{Q} = 0$, thus $|\delta F(\tau))$ remains orthogonal to $|A)$ at all time.
It also follows that

$$(\mathcal{P}\mathcal{L}A|\delta F(\tau)) = 0 \tag{2.11}$$

and

$$(\mathcal{L}A|\delta F(\tau)) = (\mathcal{L}A|\mathcal{Q}\delta F(\tau)) = (\mathcal{Q}\mathcal{L}A|\delta F(\tau)) \tag{2.12}$$

thus

$$-i(\mathcal{L}A|\delta F(\tau)) = (i\mathcal{L}A|\delta F(\tau)) = (\delta F|\delta F(\tau)) \tag{2.13}$$

substituting this back to Eq. (2.7) yields

$$e^{i\mathcal{L}t}|\delta F) = |\delta F(t)) - \int_0^t d\tau \, \frac{(\delta F|\delta F(\tau))}{(A|A)}|A(t-\tau)) \tag{2.14}$$

substuting this equation and Eq. (2.2) back into Eq. (2.1) yields

$$|\dot{A}(t)) = i\Omega|A(t)) + |\delta F(t)) - \int_0^t d\tau \, \frac{(\delta F|\delta F(\tau))}{(A|A)}|A(t-\tau)) \tag{2.15}$$

which is known as the **generalised Langevin equation (GLE)**.
We then project the entire equation onto $|A)$ to obtain

$$(A|\dot{A}(t)) = i\Omega(A|A(t)) - \int_0^t d\tau \, \frac{(\delta F|\delta F(\tau))}{(A|A)}(A|A(t-\tau)) \tag{2.16}$$

$$\dot{C}_{AA}(t) = i\Omega C_{AA}(t) - \int_0^t d\tau \, K(\tau) C_{AA}(t-\tau) \tag{2.17}$$

where we have defined the Kubo-transformed fluctuation force real time correlation function which is closely related to **memory kernel**

$$C^{\mathcal{Q}}_{\delta F \delta F}(\tau) \equiv (\delta F|\delta F(\tau)) \tag{2.18}$$

$$K(\tau) = \frac{C^{\mathcal{Q}}_{\delta F \delta F}(\tau)}{C_{AA}(0)} \tag{2.19}$$

The superscript $\mathcal{Q}$ is a constant reminder that the fluctuation force F's time evolution is governed by $e^{i\mathcal{Q}\mathcal{L}t}$ rather than the full propagator.

III

Let us define

$$\mathbf{A} = (q, p) \tag{3.1}$$

Now consider the projection

$$\mathcal{P}|U') = \frac{(q|U')}{(q|q)}|q) + \frac{(p|U')}{(p|p)}|p) \tag{3.2}$$

$$\begin{aligned}(p|U') &= \frac{1}{m}(\dot{q}|U') \\ &= \frac{1}{m}(i\mathcal{L}q|U') \\ &= -\frac{i}{m\hbar\beta}\langle[q, U'(q)]\rangle \\ &= 0\end{aligned} \tag{3.3}$$

where we have used a result from Appendix A in line 3.
and we define the mean-field frequency shift as

$$\delta\omega \equiv \frac{1}{m}\sqrt{\frac{(q|U')}{(q|q)}} = \frac{1}{m}\sqrt{\frac{C_{qU'}(0)}{C_{qq}(0)}} \tag{3.4}$$

Now consider the part nharmonic potential that is orthogonal to $|q)$:

$$(q|\delta U') = 0 \tag{3.5}$$

where

$$|\delta U') \equiv \mathcal{Q}|U') \tag{3.6}$$

we then check this using our previous definition of the fluctuating force

$$\begin{aligned}
|\delta F) &\equiv i\mathcal{Q}\mathcal{L}|q,p) \\
&= \frac{i}{\hbar}\mathcal{Q}\Big[\frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2 + U(q), (q,p)\Big] \\
&= \frac{i}{\hbar}\big[U(q), p\big] \\
&= -\mathcal{Q}|U') \\
&= -|\delta U')
\end{aligned} \tag{3.7}$$

which is expected.
Overall we have decomposed the non-linear force into two orthogonal components

$$|U') = m\delta\omega^2|q) - |\delta F) \tag{3.8}$$

which has time evolution

$$|U'(t)) = m\delta\omega^2|q(t)) - e^{i\mathcal{L}t}|\delta F) \tag{3.9}$$

using Eq. (2.14) yields

$$|U'(t)) = m\delta\omega^2|q(t)) - |\delta F(t)) + \int_0^t d\tau\, \frac{C^{\mathcal{Q}}_{\delta F\delta F}(\tau)}{C_{pp}(0)}|p(t-\tau)) \tag{3.10}$$

then project onto position operator yields

$$C_{qU'}(t) = m\delta\omega^2 C_{qq}(t) + m\int_0^t d\tau\, \frac{C^{\mathcal{Q}}_{\delta F\delta F}(\tau)}{C_{pp}(0)}\dot{C}_{qq}(t-\tau) \tag{3.11}$$

Finally substitute into Eq. (1.12) to obtain

$$\ddot{C}_{qq}(t) + \int_0^t d\tau\, \frac{C^{\mathcal{Q}}_{\delta F\delta F}(\tau)}{C_{pp}(0)}\dot{C}_{qq}(t-\tau) + \Omega^2 C_{qq}(t) = 0 \tag{3.12}$$

where we have recognised the renormlised frequency as

$$\Omega^2 = \omega^2 + \delta\omega^2 \tag{3.13}$$

Thus the the real time correlation function of position with anharmonic force contributes both to frequency shift and memory kernel.

Where the memory kernel is

$$K(\tau) = \frac{C^{\mathcal{Q}}_{\delta F \delta F}(\tau)}{C_{pp}(0)} \tag{3.14}$$

in the classical limit where equiparition is expected to hold, we have

$$K(\tau) = \frac{\beta}{m} C^{\mathcal{Q}}_{\delta F \delta F}(\tau) \tag{3.15}$$

If we let

$$K(\tau) = 2\gamma\delta(\tau) \tag{3.16}$$

then we have

$$\ddot{C}_{qq}(t) + \gamma \dot{C}_{qq}(t) + \Omega^2 C_{qq}(t) = 0 \tag{3.17}$$

which is the equation of motion of a damped oscillator.

However, if the memory kernel is in a more complicated term, we may expect a driving term to also be present.

Appendix A

Proof.

$$\begin{aligned}
\hbar Z\beta(\mathcal{L}A|B) &\equiv \int_0^\beta d\lambda \ \mathrm{tr}\Big[e^{-\beta H} e^{\lambda H} [H, A]^\dagger e^{-\lambda H} B\Big] \\
&= -\int_0^\beta d\lambda \ \mathrm{tr}\Big[e^{-\beta H} e^{\lambda H} [H, A] e^{-\lambda H} B\Big] \\
&= -\int_0^\beta d\lambda \ \mathrm{tr}\Big[e^{-\beta H} \frac{\partial}{\partial \lambda}\big(e^{\lambda H} A e^{-\lambda H}\big) B\Big] \\
&= -\mathrm{tr}\Big[A e^{-\beta H} B - e^{-\beta H} A B\Big] \\
&= \mathrm{tr}\Big[e^{-\beta H} [A, B]\Big]
\end{aligned} \tag{A1}$$

similarly

$$\begin{aligned}\hbar Z\beta(A|\mathcal{L}B) &\equiv \int_0^\beta d\lambda\ \mathrm{tr}\Big[e^{-\beta H}Ae^{-\lambda H}[H,B]e^{\lambda H}\Big] \\ &= -\int_0^\beta d\lambda\ \mathrm{tr}\Big[e^{-\beta H}A\frac{\partial}{\partial\lambda}\big(e^{-\lambda H}Be^{\lambda H}\big)\Big] \\ &= -\mathrm{tr}\Big[Ae^{-\beta H}B - e^{-\beta H}AB\Big] \\ &= \mathrm{tr}\Big[e^{-\beta H}[A,B]\Big]\end{aligned} \tag{A2}$$

thus

$$(\mathcal{L}A|B) = (A|\mathcal{L}B) \tag{A3}$$

□